THE LAWS OF SOFTWARE TEAMS

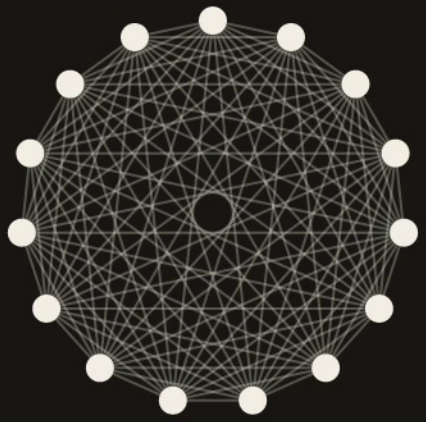
Dunbar's number

People can keep about 150 stable relationships

swipe →

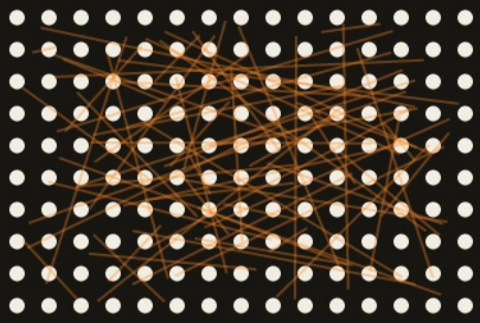
Ivan Rykov

At 15, everyone knows
At 150, nobody does



105

links · 15 people



11,175

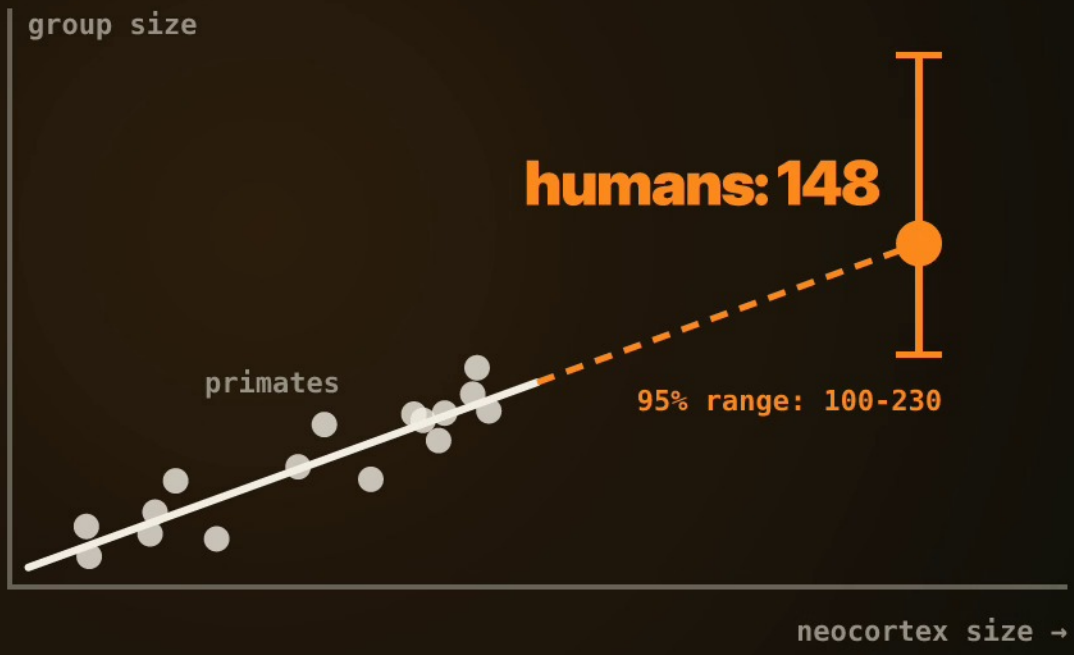
links · 150 people

`links between people = n × (n − 1) / 2`

There is a number for this

Dunbar's number: about 150

Robin Dunbar · anthropologist · 1992



Rounded to 150 stable relationships

Journal of Human Evolution, 1992 · primate points illustrative

Friends come in layers



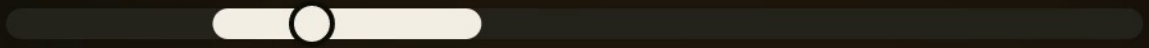
5 close · 15 good friends · 50 friends · 150

about ×3 per layer · then 500 and 1,500

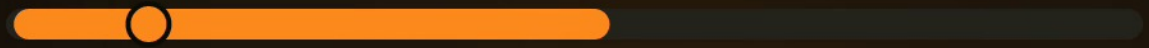
Dunbar's layers · seen again in phone-call data (MacCarron et al., 2016)

A rough idea, not a law of nature

Dunbar, 1992 — 148 · range 100-230



Re-analysis, 2021 — 69 · range 4-292



other models: averages 16-109 · widest range 5-520

0 100 200 300 400 500

LINDENFORS, WARTEL & LIND · BIOLOGY LETTERS · 2021

"A cognitive limit on human group size cannot be derived in this manner"

Companies use it as a guide

W. L. GORE & ASSOCIATES

A new plant when one grew past about 150-200 people

Gladwell, The Tipping Point (2000): 150 · company histories: about 200

SPOTIFY · 2012

“Tribes are designed to be smaller than 100 people or so”

Kniberg & Ivarsson, Scaling Agile @ Spotify

Each layer needs a new way to talk



5-9 people

a team · works by talking



~50 people

a team of teams · needs shared rituals

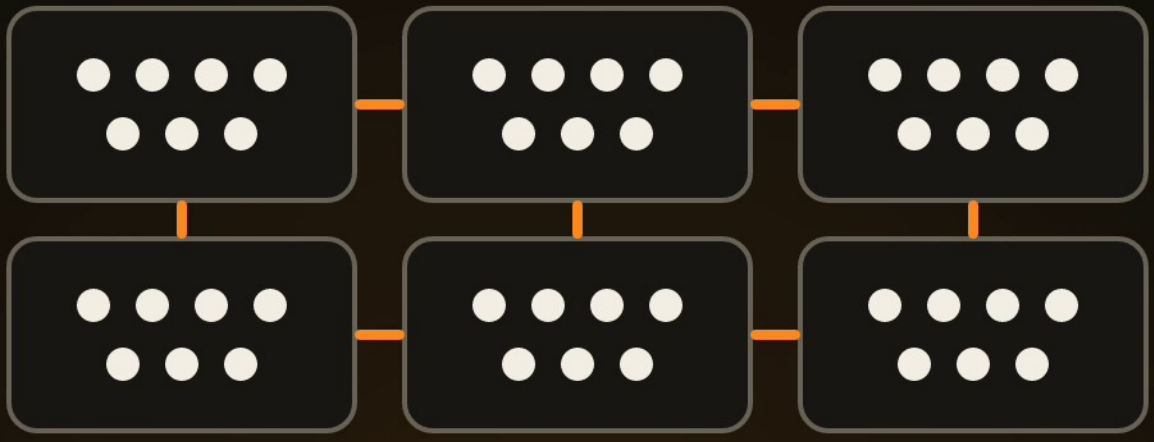


~150 people

a department · needs written decisions

a manager's reading of the layers, not Dunbar's research

Split early, along the system



Before the group stops working

CONWAY'S LAW

Team lines become system lines

illustration · amber: the interfaces between teams

08 TAKEAWAY

Grow in layers, not in one big room



Your org has a size limit, even if nobody wrote it down

TAKEAWAY

Grow in layers, not in one big room

Sources: Robin Dunbar, 1992 · Lindenfors, Wartel and Lind, 2021 · Kniberg and Ivarsson, 2012

next: the Peter principle